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Professional Summary:

- Data Engineer with 4+ years of Big Data consultant experience in Hadoop ecosystem components in **Ingestion, Data Modeling, Querying, Processing, Storage, Analysis, Data Integration and Implementing Enterprise level systems spanning Big Data.**
- Hands-on development and implementation experience on Big Data Management Platform (BMP) using **Hadoop 2.x, HDFS, MapReduce/Yarn/Spark, Hive, Pig, Oozie, Apache Nifi, Airflow, Talend, Sqoop and other Hadoop** eco-system components as Data Storage and Retrieval systems.
- Experience in designing end to end scalable architecture to solve business problems using cloud platforms **Azure Cloud, AWS and Google cloud platform (GCP).**
- Experience in designing end to end scalable architecture to solve business problems using various **Azure Services Azure - Data factory, Logic Apps, Databricks, Dataflows, EventHub, IoT Hub, Stream Analytics, Spark structured streaming, Azure DevOps.**
- Developed and optimized complex SQL queries for efficient data retrieval and storage across relational databases such as Oracle, MySQL, and Teradata.
- Designed and implemented scalable data warehouse solutions using Snowflake.
- Experience in working with AWS services **AWS Glue, Amazon Managed Kafka, Athena, AWS cloud formation templates, ECS, Network Load Balancer, API Gateway, IAM roles and policies.**
- Experience in working with GCP services **Dataflow jobs, Apache beam, Pub Sub, Cloud Composer, Apache air flow DAG for scheduling jobs, Big Query, Google Cloud Storage.**
- Experience in setting up the infrastructure for the data platform using **AWS cloud formation templates to create AWS instances and Terraform to create GCP instances.**
- Hands on experience in coding **MapReduce/Yarn** Programs using **Java, Scala and Python** for analyzing Big Data and Strong experience in building Data-pipe lines using Big Data Technologies
- Experience in Creating real-time data streaming solutions using **Spark Core, Spark SQL, Kafka, Spark Streaming, Apache Storm.**
- Experience working in Linux environments for managing and deploying data engineering solutions.
- Developed and optimized complex SQL queries in Impala to support various data processing tasks and business intelligence reports.
- Experience in importing and exporting data from various databases like **RDBMS, MYSQL, Teradata, Oracle and DB2 into HDFS using Sqoop** and also experience with different data formats like Json, Avro, parquet, RC and ORC and compressions like snappy,Gzip.
- Developed and implemented Scala-based data transformation pipelines for processing large-scale datasets in a distributed computing environment.
- Utilized Scala's functional programming features, such as higher-order functions and immutability, to create robust and maintainable data transformation code.
- Designed and implemented scalable data pipelines capable of handling terabytes of data across distributed systems.
- Configured Spark Streaming to receive real time data from Kafka and store the stream data to HDFS and process it using **Spark** and **Scala** and exposure on usage of Apache Kafka to develop data-pipe line of logs as a stream of messages using producers and consumers.
- Extensive experience working in Agile environments using both Scrum and Kanban frameworks to manage and deliver projects.
- Expertise in working with **Hive** data warehouse tool-creating tables, data distribution by implementing **partitioning** and **bucketing**, writing and optimizing the **HiveQL** queries.
- Engineered data processing workflows using Python, PySpark, and Apache Spark to handle large-scale data transformations and real-time streaming.
- Developed robust data quality checks and cleansing mechanisms to ensure the integrity and accuracy of large datasets.
- Expertize in working with internal, external tables in Hive and batch Processing jobs using Map Reduce, Hive.
- In - depth understanding of Spark Architecture including Spark Core, RDD, Data Frames, Data Sets, Spark SQL, Spark Streaming and experience in importing the data from source HDFS into Spark RDD for in-memory data computation to generate the output response.

- Hands on experience Using **Hive Tables** by **Spark**, performing transformations and Creating Data Frames on Hive tables using SparkSQL.
- Experience in converting **Hive/SQL** queries into **RDD** transformations using **Spark, Scala and Pyspark**
- Worked with Apache **Spark** components which provides fast and general engine for large data processing integrated with **Pyspark** and functional programming language **Scala**.
- Good knowledge in using apache **NiFi** to automate the data movement between different Hadoop systems.

Technical Skills:

Big Data Eco-system:	HDFS, MapReduce, Yarn, Hive, HBase, Sqoop, Kafka, Oozie, Spark, Spark SQL, Spark Streaming, Zookeeper, Cloudera, Hortonworks.
Cloud Services	Azure – ADLS, Data Factory, Databricks, EventHub, Stream Analytics GCP – Dataflow, Apache beam, Cloud Composer, Pub Sub AWS – Glue, Athena, RDS, S3, AWS cloud formation templates
Languages:	C, Java, Scala, Python, Shell Scripting
Databases:	Oracle, MySQL, Teradata
Java Technologies:	Servlets, JSP, JDBC, Spring, Hibernate, Spring MVC, Spring boot, Spring security, Spring REST
Web Technologies:	HTML, CSS, Bootstrap, JavaScript, jQuery, AJAX, Angular 2, PHP

Professional Experience:

Client: PPL Electric Utilities Corporation

Role: Data Engineer

Dec 2023 - Present

Responsibilities:

- Creating Batch Pipelines in Azure Data Factory (ADF) by configuring Linked Services/Integration Runtime to Extract, Transform and load data from different sources into Azure Data Lake.
- Creating real-time streaming pipelines using Azure IOT Hub, Azure EventHub, Azure Stream Analytics and Spark structured streaming.
- Proficient in designing and implementing ETL processes using SSIS to extract, transform, and load data from various sources into MS SQL Server databases.
- Provisioning Hadoop and Spark Clusters on the Azure Databricks, to run the spark jobs for processing the data and provide the datasets to the data scientists, data visualization teams.
- Experienced in optimizing SQL queries, indexing, and performance tuning to ensure efficient data retrieval and storage in MS SQL Server.
- Developed complex SQL queries and stored procedures in PostgreSQL to optimize data retrieval and manipulation for reporting and analytics.
- Developed and maintained ETL processes to load data into Snowflake from various sources, ensuring data integrity and quality.
- Planning and Developing roadmaps and deliverables to advance the migration of existing solutions on-premise systems/applications to Azure cloud and proposing architectures considering cost/spend in Azure and develop recommendations to right-size data infrastructure.

- Developed email notification alerts framework using Azure Logic Apps to send the emails alerts to business teams and also orchestrated azure services using Azure Logic Apps.
- Developed and implemented complex data pipelines using Apache Airflow on Azure Data Factory to orchestrate the processing and movement of large-scale big data.
- Leveraged Airflow's DAG (Directed Acyclic Graph) scheduler to automate data processing and ensure timely execution of data workflows, reducing manual intervention and improving overall project efficiency.
- Involved in analysing the different cluster types like Interactive Cluster, Job Cluster and Pool clusters and decreased the cluster usage by setting up the appropriate cluster type to the scheduled jobs.
- Conducted code reviews for peers' contributions, providing constructive feedback to ensure code quality, performance, and adherence to coding standards.
- Built the Rules for the specific Use-cases by interacting with the source team in the organization and created the extract according to the business requirement.
- Designed and executed Scala-based data quality checks and data cleansing operations to ensure the accuracy and consistency of transformed data.
- Optimized Scala transformation code for performance, leveraging techniques like partitioning, caching, and leveraging Spark's built-in optimizations for faster processing and reduced resource consumption.
- Developed and maintained secure, production-quality Python code for various data processing and transformation tasks, ensuring compliance with security standards and best practices.
- Collaborated with cross-functional teams to integrate Scala transformations into larger data processing workflows, ensuring seamless data flow and interoperability between different components.
- Programmed in PySpark, Spark SQL and Python to streamline the incoming data and build the data pipelines to get the useful insights, and orchestrated pipelines using Azure services.
- Imported data from different data sources into ADLS and created internal, external tables in Databricks which can be accessed by downstream consumers.
- Implemented continuous integration and continuous deployment (CI/CD) pipelines using GitLab, automating the deployment of code to production servers and ensuring smooth and reliable releases.
- Involved in spark performance optimizations like partitioning, bucketing and perform several types of joins on spark tables and persisted the data for the faster performance.
- Developed python scripts, UDFs using Spark-SQL, Datasets, Dataframes, RDD's in Spark for Data Aggregation
- Written generic data quality checks which are included in the data pipelines to make sure the data is flowing in different zones without any data loss or corruption.
- Involved in building automated testing framework for testing ADF pipelines, data bricks notebooks which are included in the DevOps pipeline to make the promotion into different environments.
- Involved with DevOps and Cloud teams for deploying the pipelines in different environments and also configuring the terraform scripts to create the releases in Azure DevOps for the deployment.
- Solid Understanding of Software Development Life Cycle principles.
- Used Reporting tool Power BI to connect with databricks tables for generating daily reports of data.

Environment: Azure Data Factory, Azure EventHub, Azure Logic Apps, Azure Stream Analytics, Spark Structured streaming, Databricks, Python, Spark-SQL, Azure DevOps

Client: City of New Orleans

Role: Data Engineer

July 2023 - Dec 2023

Responsibilities:

- Built ETL pipelines using AWS glue and processed the data using python, pyspark and made the data available in Athena for the ad-hoc query analysis.
- Created ECS container, Network Load Balancer, API Gateway and other AWS services using AWS cloud formation templates in Dev, QA and prod environments.
- Created Docker container and used docker images to deploy the machine learning models in different environments and automated the process using Jenkins.
- Involved in different POC's to create data pipelines in Google Cloud Platform.
- Built Dataflow jobs which ingests data from different sources, transform the data and load the data in Google Cloud Storage and created tables in Big Query.
- Designed and implemented scalable ETL processes using tools like Talend and Apache Nifi, ensuring efficient data integration from various sources.

- Leveraged AWS Step Functions to orchestrate complex workflows and automate business processes.
- Developed robust data pipelines to import and export data between RDBMS systems (MySQL, Teradata) and HDFS, optimizing performance and reliability.
- Utilized Amazon EventBridge to build event-driven architectures, facilitating seamless integration between different AWS services and custom applications.
- Managed source code and version control using GitHub, facilitating collaboration and continuous integration.
- Built ETL pipelines using AWS Glue, transforming and loading data into DynamoDB and S3 for scalable data solutions.
- Worked with Apache Beam python library to write the dataflow jobs and process the data.
- Implemented messaging and notification systems using Amazon SNS and SQS for reliable and scalable communication between services.
- Worked with different connectors in while developing the pipeline with Dataflow apache beam to develop etl pipelines and load the data in GCS and BigQuery tables.
- Automated deployment pipelines using Jenkins, ensuring seamless and reliable delivery of data engineering solutions.
- Developed serverless applications using AWS Lambda to process data streams and integrate with other AWS services like Elastic Search for real-time analytics.
- Orchestrated and automated workflows using Google cloud composer and optimized them using DAG.
- Worked with cross-platform teams to understand the business requirement and build the data pipelines according which transforms the data according to the downstream teams.
- Transformed the incoming json data by exploding the json and writing to the downstream tables according to the data scientists and visualization engineers.
- Experienced in performance tuning of Spark Applications for setting right Batch Interval time, correct level of Parallelism and memory tuning.
- Actively participated in sprint planning, daily stand-ups, sprint reviews, and retrospectives to ensure continuous improvement and timely delivery of project milestones.
- Resolved many open Spark and Yarn resource management issues, OOM errors, Shuffle exceptions, heap space errors, Null Pointer Exceptions and schema compatibility in Spark.
- handled large datasets using partitions, Spark in Memory capabilities, Broadcasts in Spark, Effective & efficient Joins, Transformation and other during ingestion process itself.
- Worked with Cloud composer to schedule the jobs using apache air flow DAG's.
- Worked along with Data Scientists to stimulate a machine learning model, hosted a python application on ECS and distributed the traffic coming from web into different servers using network load balancer.
- Configured API gateway and also applied rules to restrict public access and allowed only specific users.
- Involved in writing Terraform scripts to automate the creation of pipelines, big query tables in different environments like Dev, QA, and Prod.

Environment: AWS Glue, Athena, AWS Jupiter Notebooks, ECS Containers, Network Load Balancer, API Gateway, GCP, Data Flow Jobs, Cloud Composer, Terraform, Big Query.

Client: Palni Inc

Role: Data Engineer

Mar 2020 - Apr 2023

Responsibilities:

- Creating Pipelines in Azure Data Factory(ADF) by configuring Linked Services/Integration Runtime to Extract, Transform and load data from different sources into Azure Data Lake Store(ADLS), Azure SQL, Blob storage, Azure SQL Data warehouse, write-back tool and backwards.
- Extracting, Parsing, Cleaning and ingesting the incoming web feed data and server logs into the HDInsight and Azure DataLake Store by handling structured and unstructured data.
- Setup Databricks Enterprise Platform environment, created cross-account role in Azure for Databricks to provision Spark clusters to access Azure Data Lake Store(ADLS).
- Implemented and managed Apache Airflow workflows to orchestrate ETL processes, ensuring timely and accurate data pipelines.
- Utilized Apache Impala for high-performance querying and data analysis within the Hadoop ecosystem, improving data retrieval times and enabling real-time analytics.
- Utilized Teradata (Vantage) for advanced data warehousing solutions, performing data transformations and analytics to support business intelligence.

- Provisioning Hadoop and Spark Clusters on the Azure HDInsight, to build the On-Demand data warehouse to process the PBs of data and provide the datasets to the data scientists.
- Integrated diverse data sources into Snowflake, utilizing Snowpipe for real-time data ingestion and ensuring seamless data flow.
- Utilized Linux shell scripting for automation of data processing tasks and system monitoring.
- Programmed in Hive, Spark SQL and Python to streamline the incoming data and build the data pipelines to get the useful insights, and orchestrated pipelines using Azure Data Factory.
- Worked on ORC, Parquet file formats on HDInsight, Azure Blobs and Azure tables to store for raw data.
- Imported data from data different sources into HDFS using Sqoop, created hive internal, external tables
- Performed incremental loads using Sqoop commands and automated the process for the consistent refresh and made the data available to Data Scientist team to train their models.
- Leveraged Apache Kafka for developing real-time data streaming solutions, ensuring reliable and efficient data flow between systems.
- Conducted regular database maintenance tasks such as backups, restores, and performance tuning in PostgreSQL to ensure data integrity and availability.
- Enhanced Snowflake performance by implementing clustering, partitioning, and effective data compression techniques.
- Optimized Snowflake performance through effective partitioning, clustering, and data compression techniques.
- Developed and maintained ETL processes to load data into Snowflake from various sources, ensuring data integrity and quality.
- Developed and maintained applications using PHP, including server-side scripting and integration with databases.
- Written shell scripts and automated scripts, incremental loads, sqoop, hive, spark jobs using Oozie and cron tab schedulers.
- Involved in Hive performance optimizations like partitioning, bucketing and perform several types of joins on Hive tables and implementing Hive SerDes like JSON and Avro.
- Migrated data from other relational databases (such as MySQL and Oracle) to PostgreSQL, ensuring data consistency and integrity throughout the process.
- Developed Spark applications using Python(Pyspark) for easy hadoop transitions and used Spark API's over Hortonworks Hadoop YARN to perform analytics on data in Hive .
- Developed Spark code and Spark - SQL/Streaming for faster testing and processing of data.

Environment: Hadoop Map Reduce, Hive, Sqoop, NiFi, Teradata, HBase, Zoo Keeper, Shell Scripts, Spark Python, SparkSQL, Spark Streaming, Kafka, Oracle, IntelliJ, Azure-Data Lake, Data Factory, Data Bricks.

Education:

- Bachelor of Technology – Malla Reddy University, Hyderabad, India